

END-TO-END BIG DATA ARCHITECTURE

AIS Maritime Data Pipeline

Complete Technical Overview: Architecture, Tools Stack, Ingestion, Distributed Processing & Machine Learning

Big Data Engineering | November 2025 Dataset

Project Objective

This project aims to build an end-to-end Big Data pipeline for analyzing Automatic Identification System (AIS) data.

The system processes large-scale maritime data to provide useful insights about:

- .Ship traffic and activity
- .Vessel types

- .Speed behavior
- .Geographic distribution

- .Hourly and daily traffic patterns
- .Vessel movement clusters

Complete Technology & Tools Stack

Python & Zstandard

Pipeline orchestration, automated downloading & `.zst` file decompression

Hadoop HDFS

Distributed storage layer for raw CSV logs and partitioned Parquet datasets

Apache PySpark

Distributed data processing engine for cleaning, transformation & aggregation

PySpark MLlib

Machine Learning library for VectorAssembler & K-Means vessel clustering

Apache Parquet

Columnar storage format partitioned by date (`date=YYYY-MM-DD`)

Spark Thrift Server

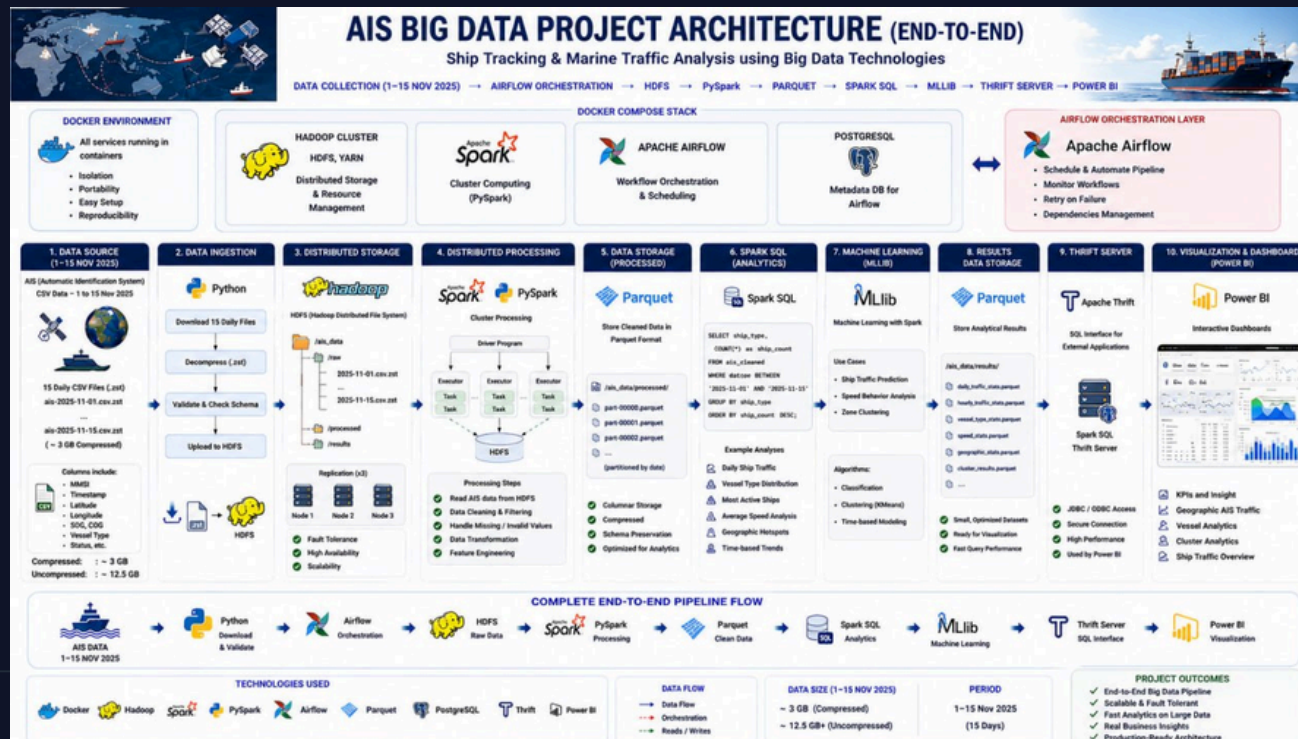
JDBC/ODBC execution engine for serving real-time SQL queries

Power BI

Interactive BI dashboard for maritime heatmaps, speed KPIs & route visualization

Airflow

Workflow orchestration



| End-to-End System Pipeline

Data Pipeline Flow

1. **Source:** AIS Raw Broadcast Logs (`.csv.zst`)
2. **Ingestion:** Python script decompresses `.zst` to `.csv`
3. **Storage:** Upload raw files to HDFS `/ais\_data/raw/`
4. **Processing:** PySpark cleans nulls, filters invalid Lat/Long
5. **ML Layer:** K-Means clustering for route identification
6. **Serving:** Hive Metastore + Spark Thrift Server
7. **Consumption:** Power BI DirectQuery / Import Dashboards

Key Pipeline Highlights

- **Fault Tolerance:** HDFS 3x block replication prevents data loss.
- **Storage Reduction:** Parquet compression reduced 12.5GB CSV to ~1.2GB.
- **High Performance:** Distributed Spark workers compute aggregations in seconds.

| Dataset Characteristics & Ingestion

- **Raw Format:** Zstandard compressed (`.csv.zst`) daily logs.
- **Timeframe:** 1–15 November 2025 (15 continuous days).
- **Data Volume:** ~3.1 GB compressed / ~12.5 GB uncompressed text.
- **Python Ingestion Engine:** Automated streaming script utilizing `zstandard` and `pathlib` for direct decompression to HDFS landing storage.
- **Scalability & Concurrency:** Structured directory partitions designed for parallel execution by downstream Spark workers.

AIS Schema & Feature Specification

Attribute	Data Type	Description & Significance	Cleaning Rule
MMSI	Long / String	Unique 9-digit vessel identifier	Drop NULL or non-9 digit IDs
BaseDateTime	Timestamp	UTC Broadcast timestamp	Parse timestamp, extract Date/Hour
LAT / LON	Double	Geographic coordinates	Filter LAT (-90 to 90), LON (-180 to 180)
SOG	Float	Speed Over Ground (Knots)	Filter SOG >= 0.0 & SOG <= 60.0
VesselType	Integer / String	Classification (Cargo, Tanker, Tug, etc.)	Map numerical codes to type labels
Status	String	Navigational status (Underway, Anchored)	Fill unknown with 'Unspecified'

| Distributed Storage Layer (HDFS)

- Hadoop Distributed File System (HDFS) is used to store the large AIS dataset.
- **HDFS Path Structure:**
 - `/ais_project/raw/` (Ingested raw CSV daily dumps)
 - `/ais_project/cleaned/parquet/` (Partitioned clean Parquet files)
 - `/ais_project/result/parquet/` (Partitioned analytic Parquet files)
- **Block Size & Replication:** 128 MB HDFS block size configured with replication factor of 3 for high fault-tolerance.
- **HDFS provides:**
Distributed storage, Fault tolerance, High availability, Scalability. The raw AIS data is stored in HDFS before processing.

PySpark Data Processing & Transformation

is used for distributed processing of the AIS data.

PySpark performs:

1. Cleaning & Quality

Removing corrupted records, invalid Lat/Long coordinates, null MMSIs, and outlier speeds (SOG > 60 knots) and Handling missing values.

2. Feature Extraction

Deriving temporal attributes (``DayOfWeek``, ``HourOfDay``, ``IsMoving``), speed categories, and spatial grid bins.

3. Parquet Conversion

Writing cleaned outputs to Snappy-compressed Parquet partitioned by date (``date=2025-11-01``).

Spark SQL Analytics

Spark SQL is used to perform analytical queries on the processed AIS data.

Examples of performed analyses include:

Daily ship traffic
Hourly traffic patterns

Vessel type distribution
Average speed

Geographic analysis
Top active vessels

The analytical results are stored in HDFS for further use.

Machine Learning Layer (PySpark MLlib)

- Apache Spark MLlib is used to perform machine-learning-based analysis.
- In our project, we implemented zone clustering based on vessel movement characteristics.
- The clustering logic uses:
 - Ship speed (SOG)
 - Geographic position (Latitude)
- The vessels are assigned to different clusters representing different movement/traffic zones.

| Serving Layer: Spark Thrift Server

Apache Spark Thrift Server

Acts as the distributed JDBC/ODBC server gateway, allowing multi-client concurrent access to Hive metastore tables defined over the Parquet files on HDFS.

Key Serving Benefits

1. **Direct JDBC Connection:** Power BI connects directly via Hive/Spark connector.
2. **Low Query Latency:** In-memory caching and Parquet column pruning yield sub-second response times.
3. **Concurrent Analytics:** Enables multiple analyst sessions simultaneously.

Visualization & Power BI Dashboard

Traffic Heatmap

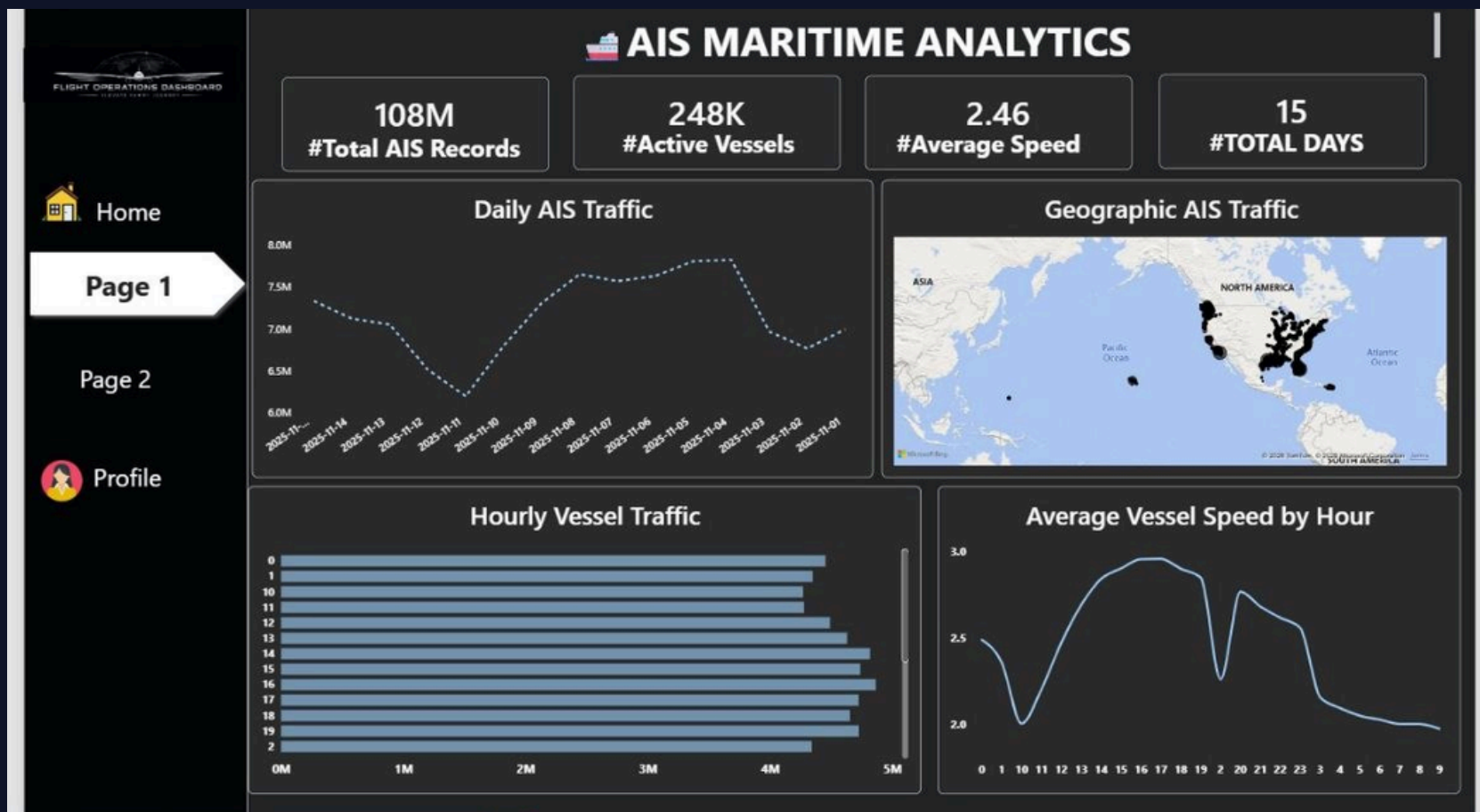
Interactive geospatial map illustrating vessel density along major coastlines and narrow straits.

Speed & Activity KPIs

Speed Over Ground (SOG) distribution bar charts by vessel category (Cargo, Tanker, Passenger).

Cluster Distribution

Visualizing MLlib K-Means clusters to highlight abnormal navigational behavior.



| Pipeline Optimization Results & Metrics

12.5GB

Raw Data Ingested

10x

Storage Savings with
Parquet

15

Continuous Ingested Days

< 2s

Average SQL Query Time

Questions & Discussion

Thank you for reviewing the AIS Big Data Pipeline project!

Tools Used: Python • HDFS • PySpark • MLlib • Spark Thrift Server • Power BI

Thank You

for your time and attention

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